title: Sir Rolly Ramos - Mechanical Engineer

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Sir Rolly Ramos

Quezon City, National Capital Region, Philippines

Professional Summary

Dedicated and detail-oriented Mechanical Engineering professional with strong knowledge in mechanical design, thermodynamics, fluid mechanics, and manufacturing processes. Experienced in using engineering software for design and analysis, with a passion for innovation, problem-solving, and continuous improvement. Committed to delivering efficient, safe, and cost-effective engineering solutions.

Education

Bachelor of Science in Mechanical Engineering

[University Name] – [City, Country]

Graduated: [Year e.g. 202X]

Relevant Coursework

- Thermodynamics
- Fluid Mechanics
- Heat Transfer
- Machine Design
- Engineering Materials
- Manufacturing Processes

- Control Systems

Technical Skills

Design & Modeling

• AutoCAD • SolidWorks • CATIA

Analysis & Simulation

• ANSYS • MATLAB

Programming

• Python (Basic) • C++ (Basic)

Domain Expertise

• Mechanical system design • HVAC systems • Hydraulic & pneumatic systems • Quality control & assurance

Professional

• Project management • Technical documentation • Root cause analysis & problem-solving

Professional Experience

Mechanical Engineer

[Company Name] – [City, Country]

[Month Year] – [Month Year / Present]

- Designed and developed mechanical components, assemblies, and systems using 3D CAD software.
- Performed structural, thermal, and fluid analysis to optimize product performance and reliability.
- Collaborated with cross-functional teams (electrical, production, quality) to meet project timelines and budgets.
- Prepared detailed technical reports, drawings, BOMs, and compliance documentation.
- Ensured adherence to safety standards (OSHA, ISO) and company quality protocols.

Engineering Intern

Department of Migrant Workers – Mandaluyong City

July 28, 2026

- Supported senior engineers in design reviews, prototyping, and equipment maintenance.

- Conducted on-site inspections, troubleshooting, and failure analysis of mechanical systems.

- Assisted in creating 2D/3D CAD drawings and updating technical documentation.

Selected Projects

Automated Conveyor System Design

- Engineered a modular, cost-effective automated conveyor for material handling in industrial settings.

- Achieved 20% improvement in throughput efficiency through optimized layout and motor selection.

Energy-Efficient HVAC System Modeling

- Designed and simulated an HVAC system focused on energy optimization.

- Reduced projected energy consumption by 15% via detailed thermal load calculations and component selection.

Certifications

- Professional Mechanical Engineer License (if applicable / PRC)

- AutoCAD Certified Professional

- Occupational Safety and Health (OSH) Training

Leadership & Extracurriculars

- Active Member, Philippine Society of Mechanical Engineers (PSME) or University Mechanical Engineering Society

- Team Leader, Capstone Design Project – [Brief project title if relevant]

- Volunteer, Community Technical Support & Repair Program

References

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[2] Barnes, M., 2001, "Stresses in Solenoids," J. Appl. Phys., **48**(5), pp. 2000–2008.

[3] Jones, J., 2000, *Contact Mechanics*, Cambridge University Press, Cambridge, UK, Chap. 6.

[4] Lee, Y., Korpela, S. A., and Horne, R. N., 1982, "Structure of Multi-Cellular Natural Convection in a Tall Vertical Annulus," *Proc. 7th International Heat Transfer Conference*, U. Grigul et al., eds., Hemisphere, Washington, DC, **2**, pp. 221–226.

[5] Hashish, M., 2000, "600 MPa Waterjet Technology Development," *High Pressure Technology*, PVP-Vol. **406**, pp. 135-140.

[6] Watson, D. W., 1997, "Thermodynamic Analysis," ASME Paper No. 97-GT-288.

[7] Tung, C. Y., 1982, "Evaporative Heat Transfer in the Contact Line of a Mixture," Ph.D. thesis, Rensselaer Polytechnic Institute, Troy, NY.

[8] Kwon, O. K., and Pletcher, R. H., 1981, "Prediction of the Incompressible Flow Over A Rearward-Facing Step," Technical Report No. HTL-26, CFD-4, Iowa State Univ., Ames, IA.

[9] Smith, R., 2002, "Conformal Lubricated Contact of Cylindrical Surfaces Involved in a Non-Steady Motion," Ph.D. thesis, http://www.cas.phys.unm.edu/rsmith/homepage.htm